

— FEATURE BRIEF · TOWER X

QS Cost — what we built

Nineteen features that help a Quantity Surveyor see cost trouble one reporting period before it lands.

19 features

39 API endpoints

173 cost centres × 12 periods

224.3M AED under management

Full detail — worked examples, formulas, honest limits — in **QS-Cost-Feature-Guide.pdf**.

— THE PROBLEM

A QS spots the overrun weeks too late.

- Two questions, forever: **what will this finally cost?** and **where are we drifting?**
- Both answered in a spreadsheet, at month-end, from numbers already weeks old.
- By the time a cost centre reads AMBER, **the money that caused it is spent.**

The line that matters

On Tower X a centre goes AMBER exactly when **CPI falls below 0.95** — the project's own rule, matching the supplied flag on every row.

The history holds **117** GREEN → AMBER transitions. Every one of them was visible the month before.

— WHAT WE BUILT

A system of record, not a report.

19

features, wired end to end
across 14 tabs

39

API endpoints over a
multi-tenant Postgres store

2,076

panel rows — every EVM figure
derived, never looked up

163

automated tests, including
the arithmetic tie-outs

The workbook's computed EVM, KPI and progress sheets were deliberately withheld. Everything this system shows for Tower X had to be computed from the bill, the estimate and 12 months of history.

Seven jobs a QS actually does.

Where we stand 2

Live EVM dashboard · cost-centre grid and detail drawer

See trouble early 4

Watchlist · variance attribution · hindsight proof · model validation

Look forward 3

Spend forecaster with a real band · unit-rate what-if · correction actions

Before you award 1

Estimate assumption stress test — three separated output classes

Ask in English 1

QS Copilot — 13 read-only tools, every answer carries its evidence

On the building 4

3D Cost X-Ray · IFC take-off · element register · 4D build sequence

Run it as a system 4

Reporting workflow · projects · data administration · tenant isolation

Every feature says what it cannot prove

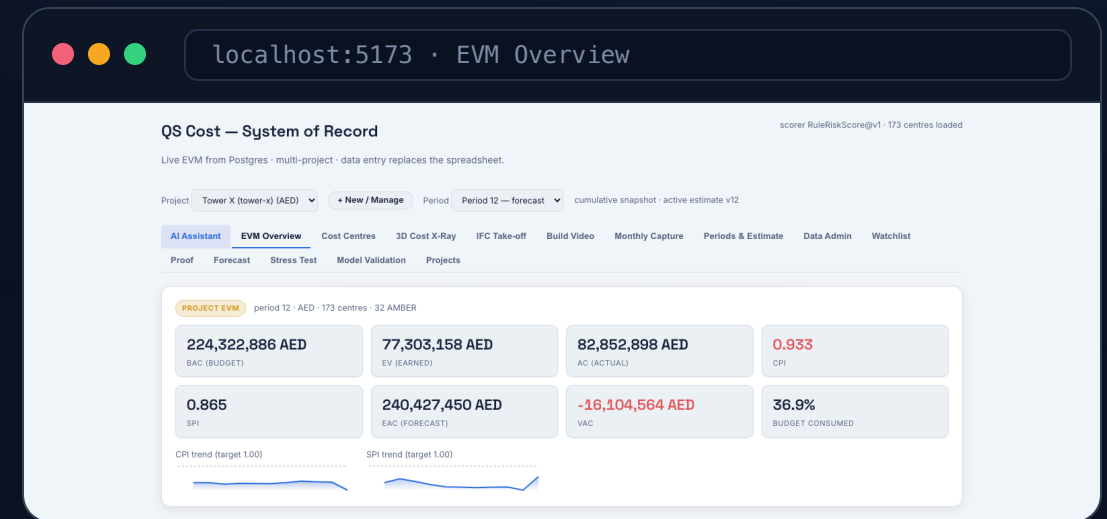
Directional figures are labelled on screen and listed in one honesty ledger at the back of the guide.

— 01 · WHERE THE PROJECT STANDS

The spreadsheet, computed.

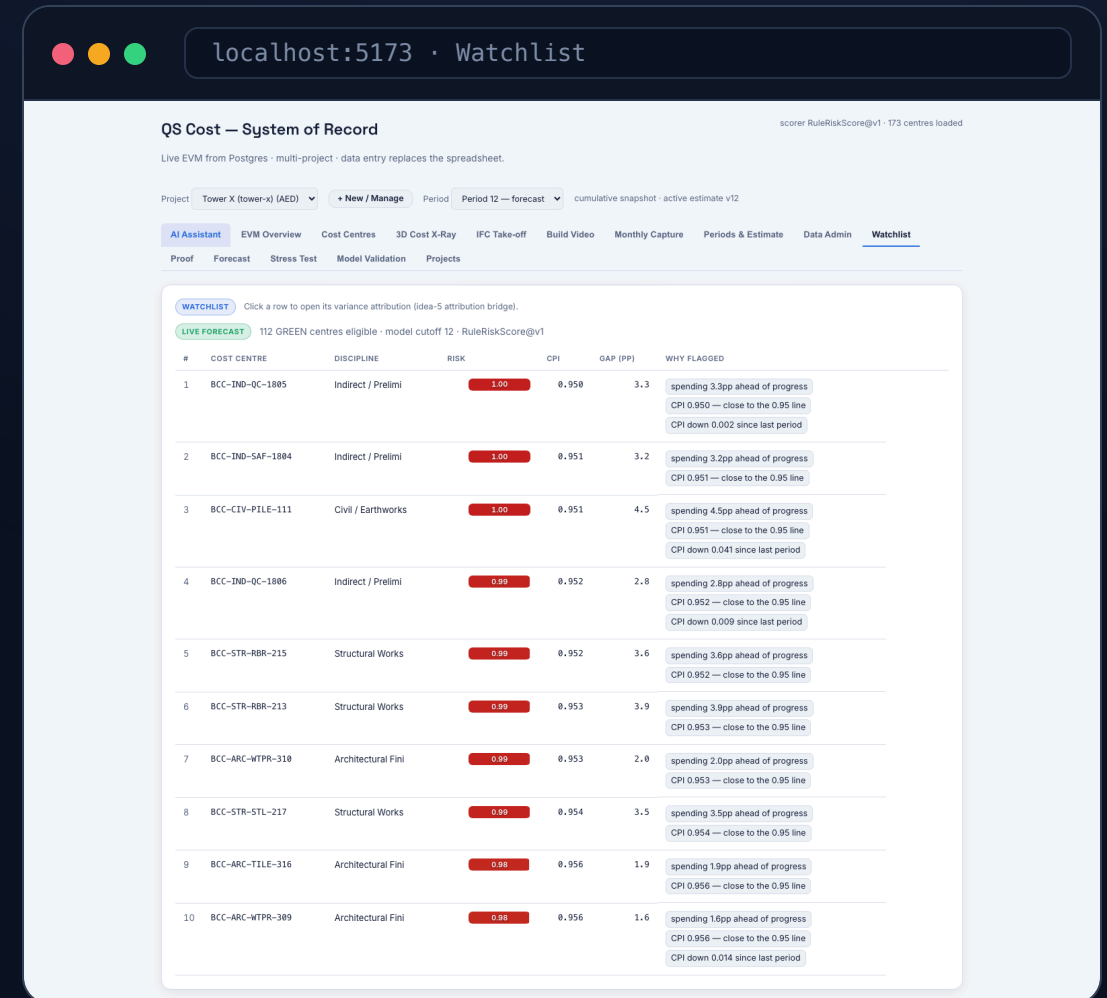
- **Live EVM Dashboard** — *what is the project's cost health right now?* Totals and CPI/SPI trends for any period, computed on request.
- **Cost Centre Grid & Detail** — *which of 173 centres owns the problem?* Sortable, filterable, one click into the full picture.

Project CPI is always $\Sigma EV \div \Sigma AC$ — never the mean of per-centre ratios.



A ranked triage list, one period early.

- **Early-Warning Watchlist** — *which GREEN centres are about to tip?* A published two-term score, not a black box.
- **Variance Attribution** — *where is the over-run coming from?* Split by resource, reconciled to the dirham.
- **Proof** — *would it have worked last month?* Rewind and let history grade it.
- **Model Validation** — *how accurate is it, honestly?*



— THE NUMBER

We let the app grade itself.

45% vs 35%

precision@5 — the deployed rule against the best CPI-native baseline

117

real GREEN → AMBER transitions, across 8 leakage-safe folds

2.8

false alerts per cycle — the cost of looking early

Published, not hidden

The per-fold range is **20%–80%** and it is on the screen next to the mean, because the mean alone would overstate how settled the number is.

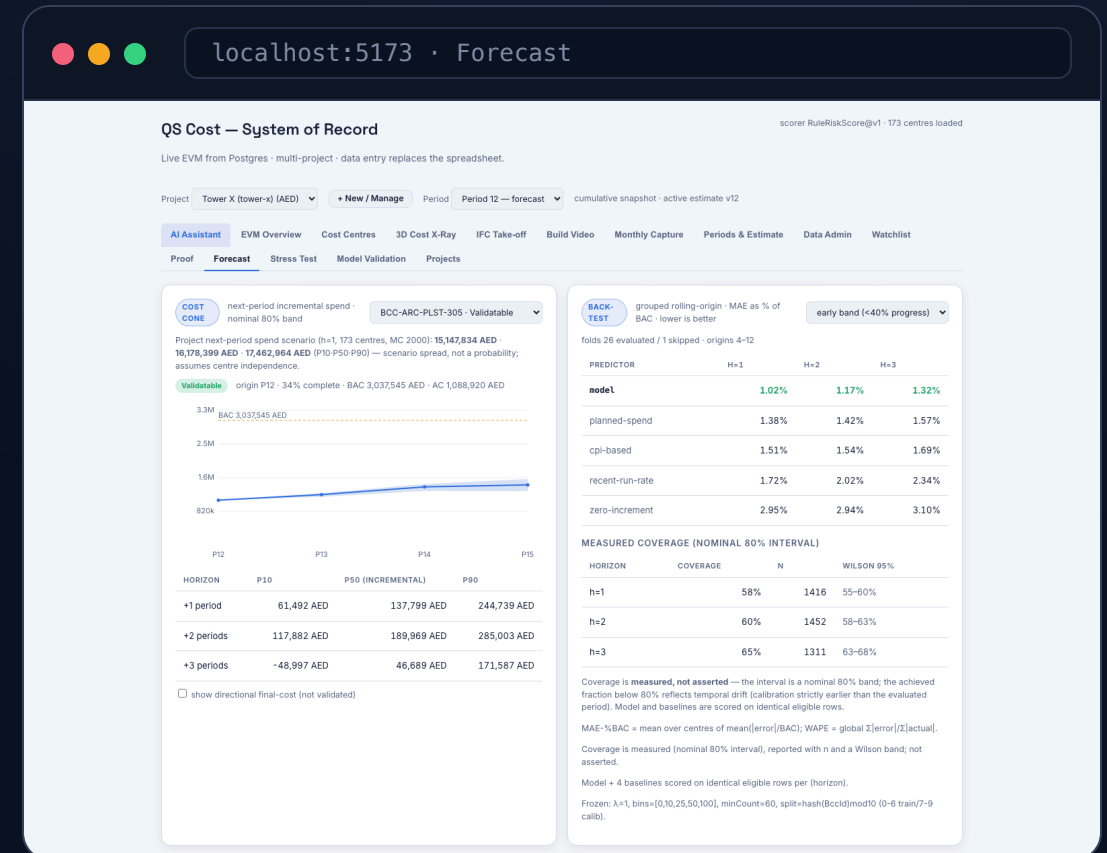
The question we could not ask

We set out to test whether physical neighbourhood predicts drift. On Tower X **no discipline spans more than one zone** — so a "spatial" feature would really measure trade. We published the finding instead of shipping the feature.

Next-period spend, with a real band.

- **Cost-Trajectory Forecaster** — *how much next period, and what range?* P10–P90 from the model's own out-of-fold errors, not an assumed distribution.
- **Unit-Rate What-If** — *what is a renegotiation worth?*
- **Correction Actions** — *what could we do, and what would it save?*

Coverage of the 80% band is **measured, not asserted**: 58% / 60% / 65% at $h=1/2/3$, published with sample sizes and confidence intervals.



Catch trouble before award.

- **Class 1 — reconciliation.** Should-cost rebuilt from norms × rates ties out to the AED across all 173 items. A correctness proof, not a signal.
- **Class 2 — assumption flags.** 38 cohort-gated review prompts: aggressive output norms, thin rates, zero contingency. Reads no actuals.
- **Class 3 — peer benchmark.** Needs 5 completed peers. Tower X has none, so it is **suppressed rather than faked.**

localhost:5173 · Stress Test

QS Cost — System of Record

scorer RuleRiskScore@v1 · 173 centres loaded

Live EVM from Postgres · multi-project · data entry replaces the spreadsheet.

Project Tower X (tower-x) (AED) · New / Manage · Period Period 12 — forecast · cumulative snapshot · active estimate v12

AI Assistant · EVM Overview · Cost Centres · 3D Cost X-Ray · IFC Take-off · Build Video · Monthly Capture · Periods & Estimate · Data Admin · Watchlist

Proof · Forecast · Stress Test · Model Validation · Projects

STRESS TEST

Estimate Assumption Stress Test · deterministic · run at award · three separated output classes

CLASS 1 — RECONCILIATION TIE-OUT

✓ Should-cost rebuilt from norms × rates ties out to the AED across all 173 BOQ items. The residual is exactly margin + contingency — this is a correctness proof, not a signal.

173
ITEMS RECONCILED

291.5M AED
CONTRACT TOTAL

49.3M AED
MARGIN

17.9M AED
CONTINGENCY

CLASS 2 — UNUSUAL ESTIMATE ASSUMPTIONS (38)

Estimate-side review prompts (read no actuals): aggressive Output Norm, thin Unit Rate, thin/zero contingency. Cohort-gated (x5), rules v1.

EP-STR-CON
Structural Works · 10

EP-FAL-BET
Fire Alarm Systems · 5

EP-COM-SEC
Communication Systems · 2

EP-KIT-EQUIP
Kitchen Works · 2

EP-LFT-SHAFT
Lifts & Escalators · 2

EP-ARC-MAS
Architectural Finishes · 1

EP-ARC-TILE
Architectural Finishes · 1

EP-BTH-ACC
Bathroom Works · 1

EP-BTH-FIN
Bathroom Works · 1

EP-BTH-SAN
Bathroom Works · 1

EP-CIV-ERTH
Civil / Earthworks · 1

EP-CLG-BULK
False Ceilings & Bulkheads · 1

EP-CLG-GRID
False Ceilings & Bulkheads · 1

EP-COM-INTER
Communication Systems · 1

EP-COM-PA
Communication Systems · 1

EP-COM-HIFI
Communication Systems · 1

EP-ELE-LV
Electrical Works · 1

EP-IND-SITE
Indirect / Preliminaries · 1

EP-INS-TILE
Insulation & Tiling System · 1

EP-KIT-FIN
Kitchen Works · 1

EP-LSC-PLANT
Landscaping · 1

EP-STR-FWK
Structural Works · 1

Kind all (38)

FLAG	PACKAGE	DISCIPLINE	DRIVING LINE	REASON
Unit rate ≤ P10	EP-ARC-MAS	Architectural Finishes	3.03 Hand tools	Unit rate 850 ≤ P10 (850) for EQUIPMENT/Hand tools (thin rate)
Unit rate ≤ P10	EP-ARC-TILE	Architectural Finishes	3.16 Tile cutter	Unit rate 850 ≤ P10 (850) for EQUIPMENT/Tile cutter (thin rate)
Unit rate ≤ P10	EP-BTH-ACC	Bathroom Works	14.05 Drill + screwdriver	Unit rate 850 ≤ P10 (850) for EQUIPMENT/Drill + screwdriver (thin rate)
Unit rate ≤ P10	EP-BTH-FIN	Bathroom Works	14.08 Tile cutter	Unit rate 850 ≤ P10 (850) for EQUIPMENT/Tile cutter (thin rate)
Unit rate ≤ P10	EP-BTH-SAN	Bathroom Works	14.03 Hand tools	Unit rate 850 ≤ P10 (850) for EQUIPMENT/Hand tools (thin rate)
Output Norm ≥ P90	EP-CIV-ERTH	Civil / Earthworks	norm CIV-ERTH-01	Output Norm 350 ≥ P90 (260) for Earthworks/m³ (aggressive productivity assumption)
Unit rate ≤ P10	EP-CLG-BULK	False Ceilings & Bulkheads	5.12 Drill + screwdriver	Unit rate 850 ≤ P10 (850) for EQUIPMENT/Drill + screwdriver (thin rate)
Unit rate ≤ P10	EP-CLG-GRID	False Ceilings & Bulkheads	5.05 Tile cutter	Unit rate 850 ≤ P10 (850) for EQUIPMENT/Tile cutter (thin rate)
Unit rate ≤ P10	EP-COM-PA	Communication Systems	13.18 Drill + screwdriver	Unit rate 850 ≤ P10 (850) for EQUIPMENT/Drill + screwdriver (thin rate)

Tools compute. The model **narrates**.

- 13 read-only tools running the same C# analytics the dashboard uses — the model has no database access and does no arithmetic.
- Every answer carries a **sources block** naming the tools called and the rows read.
- Tenancy is resolved **before** the model runs, so no prompt can reach another project.

Validated-versus-directional is enforced in the tools, not the prompt: the final-cost tool returns its "not validated" flag as part of the payload.

The screenshot shows a web application titled "QS Cost — System of Record" with a subtitle "Live EVM from Postgres · multi-project · data entry replaces the spreadsheet." The interface includes a top navigation bar with "localhost:5173 · AI Assistant" and a main content area with a sidebar menu. The sidebar menu has sections for "AI Assistant" (with sub-items: Proof, Forecast, Stress Test, Model Validation, Projects) and "EVM Overview" (with sub-items: Cost Centres, 3D Cost X-Ray, IFC Take-off, Build Video, Monthly Capture, Periods & Estimate, Data Admin, Watchlist). The main content area displays a "QS Cost Copilot" chat interface. The chat header says "Ask in plain English — every number is read through tested tools and shows the rows behind it." A blue button asks "Which centres are drifting this period, and why?". The chat message shows a filter for "Period 12" and a list of 10 centres at risk of tipping to AMBER. The list includes columns for Rank, Centre, Discipline, CPI, Gap, and Why. The chat interface also has a footer with "Ask about a centre, forecast, or the project CPI..." and an "Ask" button. At the bottom, a "DRIFT WATCHLIST" section shows "5 centres flagged this period — expand to view".

QS Cost — System of Record

Live EVM from Postgres · multi-project · data entry replaces the spreadsheet.

Project: Tower X (tower-x) (AED) · New / Manage · Period: Period 12 — forecast · cumulative snapshot · active estimate v12

AI Assistant · EVM Overview · Cost Centres · 3D Cost X-Ray · IFC Take-off · Build Video · Monthly Capture · Periods & Estimate · Data Admin · Watchlist

Proof · Forecast · Stress Test · Model Validation · Projects

QS Cost Copilot

Ask in plain English — every number is read through tested tools and shows the rows behind it.

Which centres are drifting this period, and why?

GREEN centres most at risk of tipping to AMBER — Period 12 (live forecast)

Filter: **Period 12** (live forecast), top 10 of **112** scoreable GREEN centres**.

Rank	Centre	Discipline	CPI	Gap (spend ahead of progress)	Why
1	BCC-IND-QC-1805	Indirect / Preliminaries	0.950	3.3pp	Sitting right on the 0.95 AMBER line, spending 3.3pp ahead of physical progress, and CPI slipping (-0.002 vs last period)
2	BCC-IND-SAF-1804	Indirect / Preliminaries	0.951	3.2pp	CPI just above the line, 3.2pp cost-ahead-of-progress gap
3	BCC-CIV-PILE-111	Civil / Earthworks	0.951	4.5pp	Largest gap in the top 10 (4.5pp) plus a sharp CPI drop (-0.041 since last period) — fastest-moving toward AMBER
4	BCC-IND-QC-1806	Indirect / Preliminaries	0.952	2.8pp	Close to the line, CPI down 0.009
5	BCC-STR-RBR-215	Structural Works	0.952	3.65pp	Spending 3.6pp ahead of progress, CPI near threshold
6	BCC-STR-RBR-213	Structural Works	0.953	3.86pp	Similar rebar-package pattern — cost outrunning progress
7	BCC-ARC-WTPR-310	Architectural Finishes	0.953	2.0pp	Smaller gap but CPI still under the 0.95+margin band
8	BCC-STR-STL-217	Structural Works	0.954	3.47pp	Steel package tracking the same structural-works drift pattern
9	BCC-ARC-TILE-316	Architectural Finishes	0.956	1.93pp	Mildest of the group, but still flagged
10	BCC-ARC-WTPR-309	Architectural Finishes	0.956	1.56pp	CPI down 0.014 since last period — moving the wrong direction

Pattern: the common driver across all ten is the same signature — **CPI sitting just above the 0.95 AMBER line while actual spend has outpaced physical progress (the "gap" column)**. Structural Works (rebar/steel) and Indirect/Preliminaries clusters dominate the top ranks, with BCC-CIV-PILE-111 the standout for both size of gap (4.5pp) and speed of CPI decline (-0.041 in one period) — worth checking first if you want to subvert the cost center.

Ask about a centre, forecast, or the project CPI...

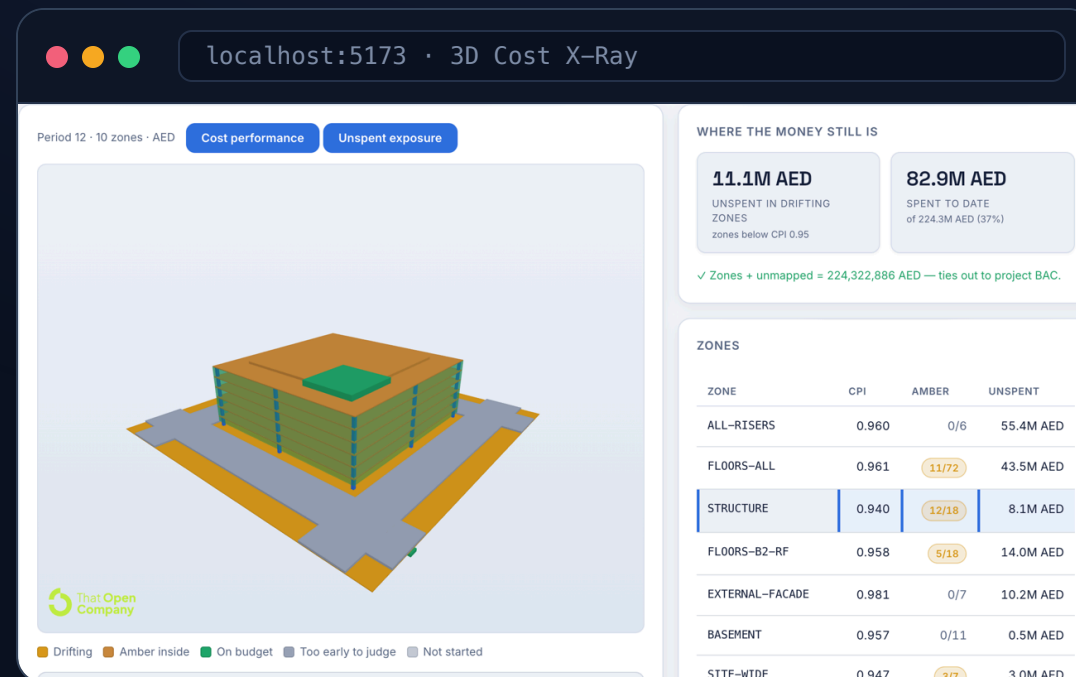
Ask

DRIFT WATCHLIST · 5 centres flagged this period — expand to view

The watchlist, on the building.

- Massing **derived from the bill** — every dimension carries the BOQ line it came from.
- **11.1M AED still unspent** in zones already below CPI 0.95.
- Zones + unmapped **ties out to project BAC**, exactly. Unlocated centres are shown, not dropped.

STRUCTURE has the worst ratio at CPI 0.940. FLOORS-ALL looks healthier at 0.961 — and has 43.5M AED still to lose.



Point it at **any model**. Get a price.

- **IFC Take-off** — a real Revit export, measured and priced with Tower X's rate library: **4.64M AED**, and **508 priced + 375 unpriced + 643 unmeasured = 1,526** elements.
- **Element register** — click an element, read the bill off it. 1,127 of 1,526 mapped, reaching 8 cost centres.
- **4D build sequence** — the sheet's progress curves, played on the model, coloured by alert. Rendered deterministically, byte-identical every run.

The 375 unpriced beams are the point: the bill prices no beam concrete, so they stay grey ghosts. The scope gap is visible as structure that never fills in.

school_str.ifc Load another IFC

PRICED AT TOWER X'S RATES

4.6M AED
PRICEABLE SCOPE
4,638,842 AED

58%
MEASURABLE
883 of 1526 elements

✓ 508 priced + 375 unpriced + 643 unmeasured = 1526 elements in the model.

WHAT COULD BE PRICED

CLASS	QUANTITY	RATE	AMOUNT
IFCSLAB · 2.06	2,735.5 m³	1122.59	3,070,815 AED
IFCSLAB · 2.11	6,761.8 m³	181.80	1,229,315 AED
IFCWALL · 2.05	127 m³	1459.66	185,443 AED
IFCCOLUMN · 2.04	112.3 m³	1364.28	153,268 AED

This is not Tower X. It is a school's structural model (Autodesk Revit sample, IFC4) being priced with Tower X's rate library. The two buildings are unrelated — what is being demonstrated is that a rate library travels to...

Production-shaped, not a notebook.

Reporting workflow 16

Open and close periods, capture progress and cost as idempotent ledger deltas, publish estimate versions, rebaseline, cut over. Enforced by database procedures, not just the API.

Project management 17

Create or import a project from a workbook in one upload, with a reconciliation summary you can check. Re-import, rename, delete.

Data administration 18

Governed CRUD over 13 entities, grouped the way the source workbook is. The forms are generated from a server registry, so the screen cannot drift from what the server accepts.

Tenant isolation 19

PostgreSQL row-level security — the database refuses another tenant's rows even if the application asks. Authorisation runs before the Copilot's tools are built.

Honest where it counts.

Validated

- Incremental spend — back-tested against four baselines on identical rows
- Estimate reconciliation — ties to the dirham, asserted on every build
- Variance tie-out — $\sum CV_r + \text{residual} = CV$, exactly
- Take-off element count and the byte-identical video render

Directional — and labelled on screen

- Final cost / EAC — no ground truth at 13% median progress
- Variance attribution — uses estimate shares, names the evidence needed
- The IFC model is a school, not Tower X — mechanism, not valuation
- 4D build order is assumed; authentication is a development shim

Fourteen such claims, each with its reason, are listed in one place — the honesty ledger at the back of the feature guide.

— UNDER THE HOOD

Four layers, one arithmetic.

Store

PostgreSQL 15

Raw SQL schema, forced row-level security, security-invoker views, stored procedures for the reporting workflow. EVM identities computed in the database.

Analytics

C# / .NET 8

Risk scorer, spend forecaster, stress tester, variance attributor, take-off pricer. Ridge regression, conformal intervals and the Cholesky solve are hand-written — **no ML dependency, no Python at run time.**

API

ASP.NET Core

39 endpoints, 13 controllers. Every read of project data is authorised before it runs.

Interface

React 18 · three.js

14 tabs, hand-built SVG charts, a WebGL viewer over real IFC, and a Copilot on Claude Sonnet through the Microsoft Agent Framework.

— READ THE DETAIL

Nineteen features, one question each.

The feature guide gives every one of them a worked example from Tower X, the endpoint behind it, and what it cannot prove.

QS-Cost-Feature-Guide.pdf — 68 pages

tower-4d-build.mp4 — the build, played

docs/ — engineering deep dives